EE10510: Fundamentals of Electrical & Electronics Engineering
Practice Sheet-II
Numerical Problems on Three-Phase Circuits

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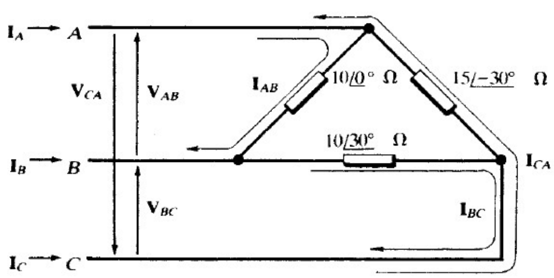
1. Three 100 ohm non inductive resistance connected in (i)star and (ii)delta connection across a 400V, 50 Hz supply. (a) calculate power taken form supply in each case (b) calculate power taken form supply when one of the resistance is open circuited. **(Ans.1600W,4800W,800W,3200W)**
2. A three phase star connected RL load has 10Ω resistance and 31.83mH inductor in series per phase is fed by a star connected sinusoidal voltage source of voltage 141.42V/ph, 50Hz. Calculate the (a) load phase voltage and current (b) load line voltage and current (c) power factor (d) Active and reactive power per phase (e) three phase apparent power. **(Ans. 141.42 $\angle 0°$ V, 10 $\angle$-45° A. 244.5 $\angle$30° V, 10 $\angle$-45° A, 0.707, 1000 W, 1000 VAR, 4242.6 VA.)**
3. A three phase delta connected RL load has 10Ω resistance and 31.83mH inductor in series per phase is fed by a star connected sinusoidal voltage source of voltage 141.42V/ph, 50Hz. Calculate the (a) load phase voltage and current (b) load line voltage and current (c) power factor (d) Active and reactive power per phase (e) three phase apparent power. **(Ans 244.5 $\angle$30° V, 17.32 $\angle$-15° A. 244.5 $\angle$30° V, 30 $\angle$15° A, 0.707. 2994 W, 2994 VAR, 12704.22 VA)**
4. A three phase star connected RL load has 10Ω resistance and 31.83mH inductor in series per phase is fed by a delta connected sinusoidal voltage source of voltage 245V/phase and 50Hz. Calculate the (a) load phase voltage and current (b) load line voltage and current (c) power factor (d) Active and reactive power per phase (e) three phase apparent power**.(Ans. 141.42 $\angle$-0° V, 10 $\angle$-45° A, 244.5 $\angle$30° V, 10 $\angle$-45° A, 0.707, 1000 W, 1000 VAR, 4242.6 VA,)**
5. A three phase delta connected RL load has 10Ω resistance and 31.83mH inductor in series per phase is fed by a star connected sinusoidal voltage source of voltage 282.4V (L-L) and 50Hz. Calculate the (a) load phase voltage and current (b) load line voltage and current (c) power factor (d) Active and reactive power per phase (e) three phase apparent power. **(Ans. 282.4 $\angle$30° V, 19.97 $\angle$-15°,282.4 $\angle$30° V, 34.6 $\angle$15° , 0.707, 3987.7 W, 3987.7 VAR, 16918.6 VA.)**
6. A three phase star connected RL load has 10Ω resistance and 31.83mH inductor in series per phase is fed by a star connected sinusoidal voltage source of voltage 141.42V/phase and 50Hz through a transmission line having impedance of (1+j1)Ω/line. Calculate the (a) load phase voltage and current (b) load line voltage and current (c) power factor (d) Active and reactive power per phase (e) three phase apparent power.
   (Ans. (a)128.06$\angle 0^o$, 128.06$\angle 120^o$, 128.06$\angle 240^o$, 9.07$\angle 45^o$, 9.07$\angle 75^o$, 9.07$\angle 195^o$, (b) 221.79$\angle 30^o$, 221.79$\angle 150^o$, 221.79$\angle 270^o$,(c) 0.707(d) 821W, 821VAR (e) 10.45KVA)
7. A three phase 339.4V has a delta connected load with Z_{AB} = 10$\angle$0° Ω Z_{BC} = 10$\angle$30° Ω and Z_{CA}= 15$\angle$-30° Ω. Calculate the (a) load phase current (b) load line current (c) Active and reactive power per phase (d) three phase apparent power.



[Ans. 33.94$\angle$120A, 33.94$\angle$-30A, 22.63$\angle$270A,54.72$\angle$108.1A, 65.56$\angle$45A, 29.93$\angle$-161.1A,11.51kW,9.91kW, 6.651kW, 0 KVA,5.75KVA, 3.84kVA, 45.061KVA]

8. A three phase four wire 150V, feeding a Y-Connected load, with Z_A = 6$\angle$0° Ω Z_B = 6$\angle$30° Ω and Z_C= 5$\angle$45° Ω. Calculate the (a) load phase current (b) load line current (c) Active and reactive power per phase (d) three phase apparent power. **(Ans. I_a=14.43$\angle$-90^0,I_b=14.43A,I_c=17.32$\angle$105^0,I_n=10.21$\angle$-167^0)**
9. A three phase three wire 150V, feeding a Y-Connected load, with Z_A = 6$\angle$0° Ω Z_B = 6$\angle$30° Ω and Z_C= 15$\angle$45° Ω. Calculate the (a) load phase current (b)displacement neutral voltage, $V_{ON.}$ **(Ans.I_a=11.13$\angle$-2.85^0,I_b=16.78$\angle$-98.92,I_c=19.12$\angle$116.4^0, V_{oN}=20.24$\angle$39.543^0)**